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ECE 3300

4/6/2026

The use of Wi-Fi matching is one we use every single day. In fact, you are using it right now if you are using your computer at home or school. This was introduced in 1997 and utilized a wireless network based on the IEEE802.11 standards. The data is transferred via electromagnetic with 3 standard wavelengths. These are 2.4 GHz, 5GHz, and 6GHz. For many years the 2.4 GHz and 5GHz were the main transport. The need for Wi-Fi connections has grown exponentially in the last decades as more and more devices use it to connect to the internet and to allow us to do multiple tasks. Then how does this really work? The Wi-Fi network comes down to 3 main components. These are the modem, access point and the client's device.

The project that for the class focuses on this Wi-Fi concept. The need to design a Wi-Fi antenna feed matching using a 2.4GHz network. The specification for this project is to construct a quarter-wave impedance transformer to match a $100\ \Omega$ antenna feed to a $50\ \Omega$ RF source. The PCB was created using the substrate RO4350B with a Dielectric constant: $\epsilon_r = 3.48$, Loss tangent: 0.0037, Substrate thickness of 1.56 mm.

The first step into creating the board is to determine the values needed to build the board in Ansys. We have been given some value in the problem. These values are $Z_l = 100\ \Omega$, $Z_o = 50\ \Omega$, $f = 2.4$ GHz. To calculate the transformer impedance, we calculated the square root of $(Z_o)(Z_l)$. This gives us an answer of $70.7\ \Omega$. The start of the design on the Ansys needs to start on the left-hand side with a $100\ \Omega$ input signal, then exit with the $50\ \Omega$ on the right-hand side. The need to calculate width for the microstrip needed to be calculated. This was calculated using the provided module 2.3 in the book modules on Moodle. The $100\ \Omega$ impedance (figure #1) had calculations showing width(W) = 0.89459 mm, height (h) = 1.56mm, frequency = 2.4×10^9 Hz, $\epsilon_r = 3.48$. This provided an $E_{eff} = 2.5199$. Then needed to calculate the values of the $50\ \Omega$ impedance. This provided values of $W = 3.54705$ mm, height = 1.56mm, frequency = 2.4×10^9 Hz, $\epsilon_r = 3.48$, and $E_{eff} = 2.7549$. Then needed to calculate $70.7\ \Omega$ which is the transmission impedance. This provided $W = 1.92973$ mm, $h = 1.56$ mm, $f = 2.4 \times 10^9$ Hz, $\epsilon_r = 3.48$, and $E_{eff} = 2.63034$. Using the formula of $\lambda = V_p/f$, it is calculated that the frequency is 0.771m. The next was calculated $\lambda/4 = 0.771\text{m}/4 = 19.275\text{mm}$. This shows that the quarter wave transformer wavelength is 19.275mm. This data was then transferred into Ansys to build the circuit. (Figure #4) The is calculated

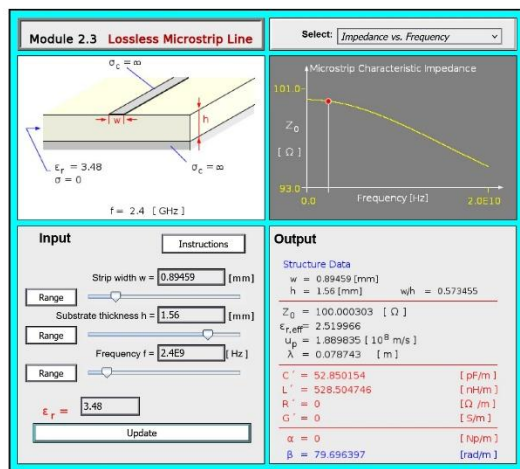


Figure #1

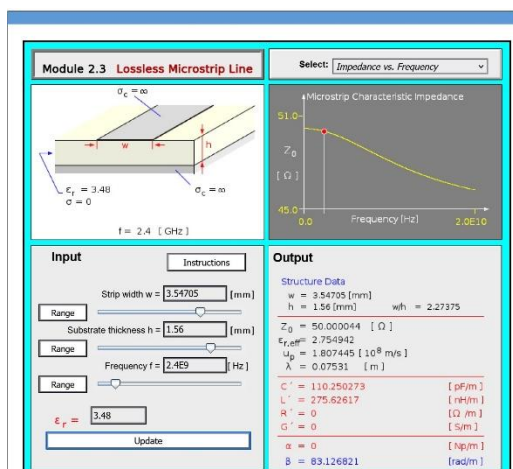


Figure #2

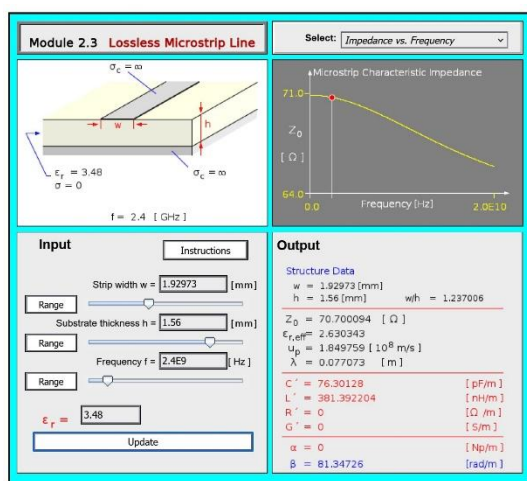


Figure #3

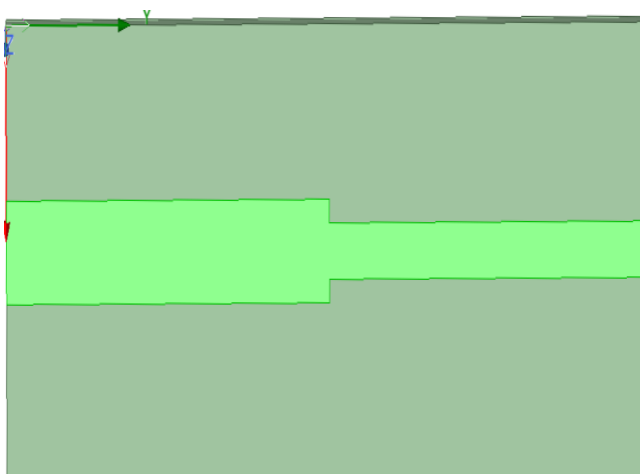


Figure #4

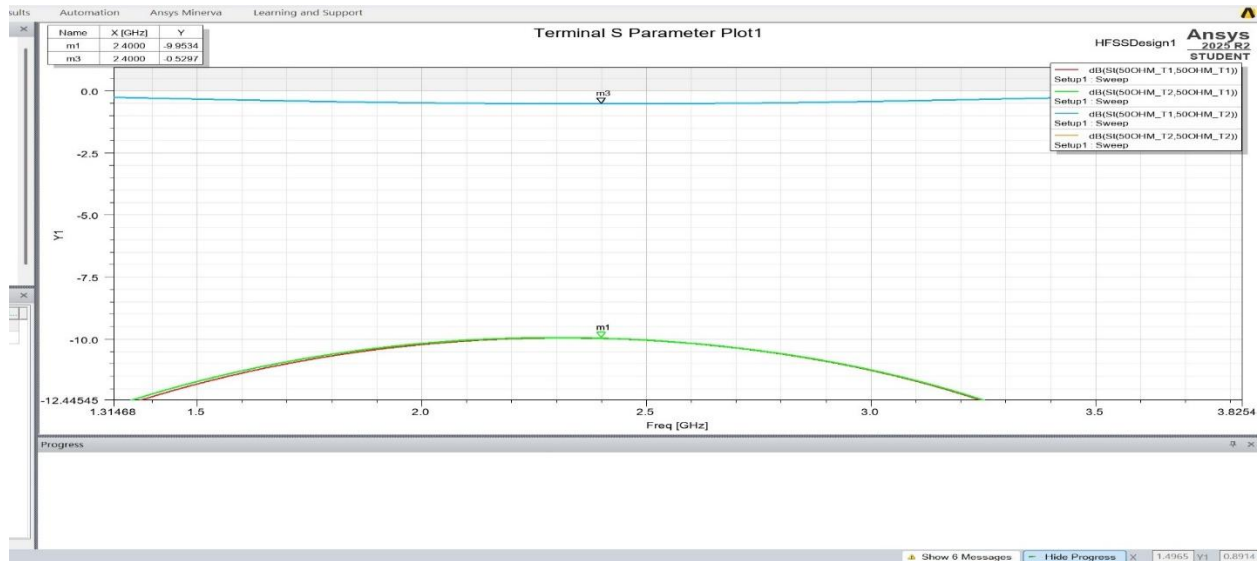


Figure #5

The Ansys data was sent into Ansys and built the circuit, and a chart was created above (Figure #5).

The next step of the project was to change the geometry in the HFSS. The way that it was changed was to adjust the width of the $\lambda/4$ value. I decided to decrease the width of this value to see what the change has shown. I also decided to change the $\lambda/4$ value to be the same width as the 50 Ω value on the left side of the board. The first adjustment was to decrease the width of the $\lambda/4$ area from 1.929 to 1.00. This provided us with a change of the 2.4 GHz values to -5.3947, and -1.5846. (Figure #6) (Figure #7)

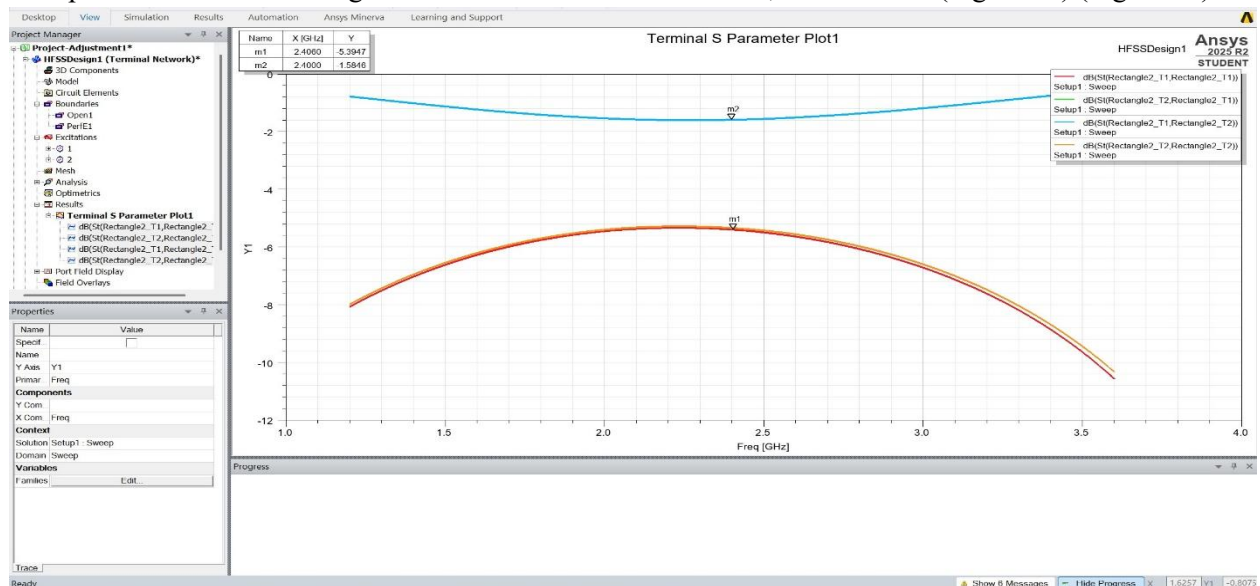


Figure #6

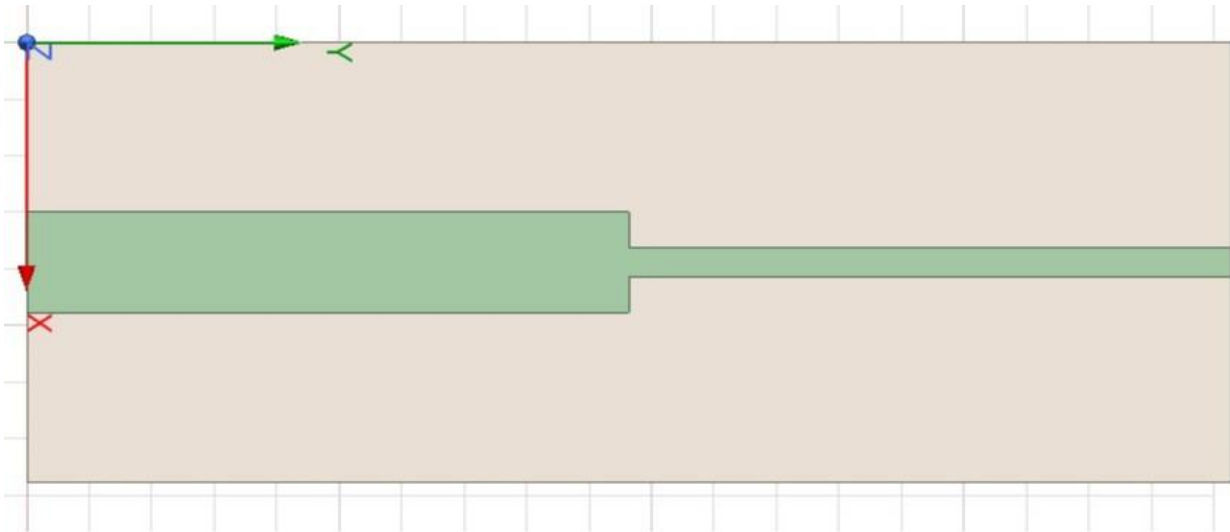


Figure #7

For the second adjustment I decided to adjust the width of the $\lambda/4$ to 3.53, which is nearly the same width as the $50\ \Omega$ section. We see that the values at 2.4 GHz were -39.1495, and -36.2230(Figure #8)(Figure #9)

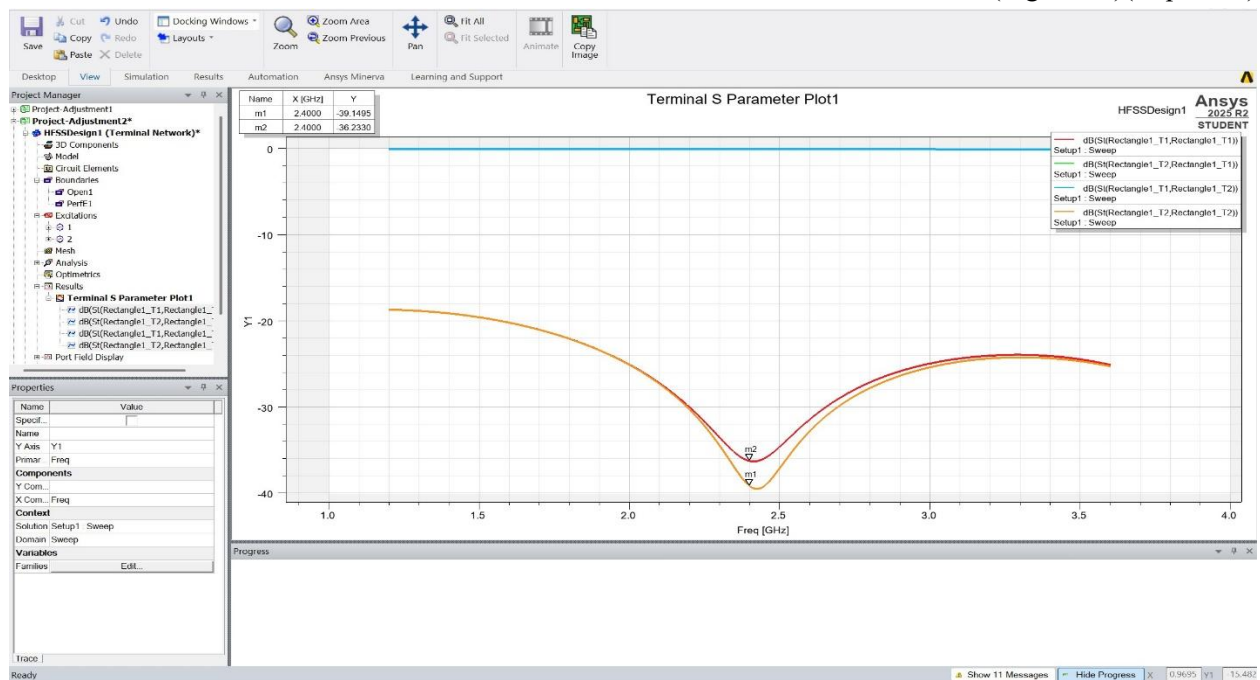


Figure #8

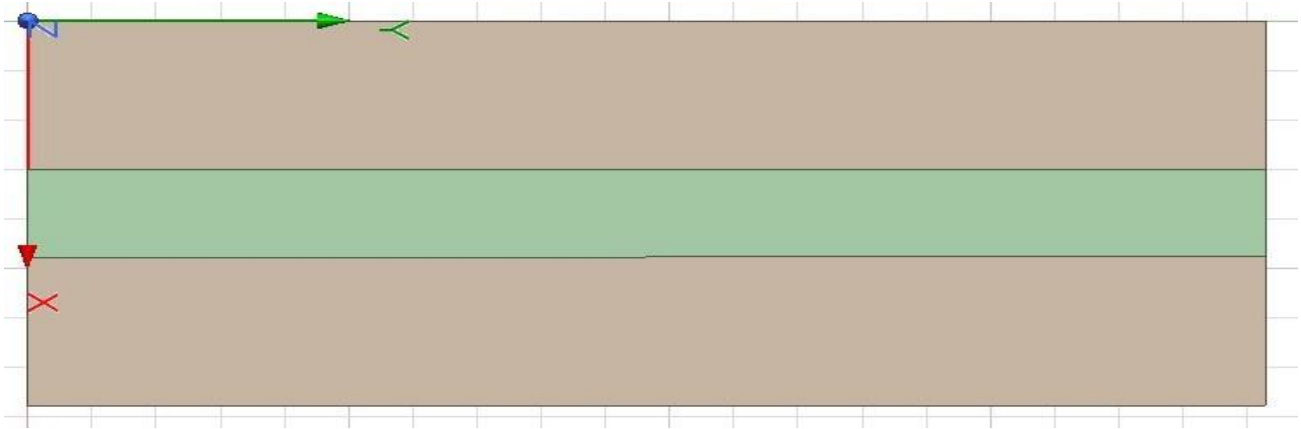


Figure #9

Research

Quarter-wave impedance transformers are used in industry in some important areas excluding the Wi-Fi area. These are antenna circuits and power circuits. Printed circuit boards and medical ultrasonic transducers. These devices can be used as a component in a filter, and in this application, it is sometimes known as an inverter because it produces the mathematical inverse of an impedance. A transmission line that is terminated in some impedance, Z_L , that is different from the characteristic impedance, Z_0 , will result in a wave being reflected from the termination back to the source. In antenna systems, a particular kind of transmission line section called a quarter-wave transformer is used to match the impedance of an antenna to that of a transmission line. Impedance matching is essential for effective power transfer, reducing signal reflections, and maximizing antenna system performance. One quarter wavelength is conveniently short enough to incorporate the component within many products at radio frequencies of upper VHF or higher up to microwave frequencies, but not so small that it cannot be manufactured using standard engineering tolerances. The device is most frequently encountered at these frequencies. It is particularly helpful when converting a capacitor into an inductor because designers favor the latter.

References

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- Fawwaz Tayssir Ulaby, E. Michielssen, and Umberto Ravaioli, *Fundamentals of Applied Electromagnetics*. PEARSON, 2010.
- GeeksforGeeks, "Quarter Wave Transformer," *GeeksforGeeks*, Jan. 03, 2024.
<https://www.geeksforgeeks.org/electrical-engineering/quarter-wave-transformer/>